AZIZ ASOMIDDINOV

Software Engineer

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SUMMARY

Software Engineer with 5+ years of experience building scalable, high-impact systems across enterprise and product environments. At General Motors, owned three mission-critical IT automation services that saved the company over $1.5M annually and reduced infrastructure provisioning timelines from weeks to days. Currently at Sordar Studios, architecting backend systems for commercially shipped indie games played by 10,000+ users. Graduated with M.S. in Data Science from the University of Denver, where research contributions included deep learning models trained on 20,000+ image datasets for medical and environmental applications.

AREA OF EXPERTISE

- **Languages:** Java, Python, JavaScript/TypeScript, C/C++, Golang, SQL, HTML5, CSS/SCSS
- **Backend:** Spring Boot, Flask, RESTful APIs, JPA/Hibernate, Authentication & Authorization
- **Frontend:** React, Angular, Responsive UI, Bootstrap, SCSS, React Router, Algolia
- **Databases:** PostgreSQL, MySQL, MongoDB, SQL/NoSQL modeling, CRUD optimization
- **DevOps & Cloud:** Azure DevOps, Docker, Kubernetes, CI/CD pipelines, AWS, Google Firebase, PCF, Netlify
- **AI Tools & Developer Productivity:** Claude Code, GitHub Copilot, Cursor, ChatGPT/Codex
- **AI/ML & Data:** TensorFlow, Keras, PyTorch, Scikit-learn, CNNs, Transfer Learning, Pandas, NumPy, XGBoost
- **Testing & Tools:** JUnit, Mockito, Selenium, Postman, Git/GitLab, Jira, Agile/Scrum

EXPERIENCE

Fullstack Software Engineer

Sordar Studios

September 2023 - Present | Remote

*Indie game studio (2–10 people) that has shipped **House of Empty Names** and **Echoes of the House** — currently in pre-production on a new co-op title.*

- Built and maintained backend systems serving **10,000+ players** across two commercially shipped titles, handling player accounts, game state, leaderboards, achievements, and live content updates.
- Architected **Flask-based RESTful and real-time APIs** that collect in-game telemetry and power analytics dashboards for both players (stats, progression) and developers (behavioral data, performance monitoring, live ops).
- Designed and optimized database schemas across relational and NoSQL stores to support high-concurrency, low-latency gameplay — managing game state, matchmaking, and telemetry pipelines at scale.
- Built responsive React/TypeScript frontends integrating analytics dashboards and player-facing features directly with backend APIs and game engines.
- Implemented authentication, session management, and security hardening to protect player accounts and defend against common online game exploits.
- Leveraged AI-assisted development tools (Claude Code, GitHub Copilot, Cursor) to accelerate feature delivery and maintain code quality across a lean team.
- Automated build, test, and deployment workflows with CI/CD pipelines; containerized services with Docker for consistent, scalable cloud deployments.

Software Engineer

General Motors

July 2021 - August 2023 Austin, TX

Joined as a college new hire on the internal IT Platform team providing infrastructure automation services to enterprise teams across General Motors.

- Solely owned and operated three mission-critical platform services — **Load Balancer, Firewall, and Server Decommissioning** — collectively saving General Motors **over $1.5M per year** in operational costs.
- Reduced infrastructure provisioning timelines from **weeks to days** by enhancing automation logic across owned services, eliminating manual handoffs and error-prone processes that previously bottlenecked enterprise IT teams.
- Contributed to a platform of **20+ internal services and APIs** built with Java Spring Boot, taking full end-to-end ownership (backend logic, frontend interfaces, monitoring, and CI/CD) for 3 of those services.
- Integrated backend services with PostgreSQL, MySQL, and MongoDB via JPA/Hibernate; implemented optimized transactional workflows to support high-volume infrastructure automation at enterprise scale.
- Built and maintained CI/CD pipelines in Azure DevOps with automated testing, security scanning, and blue-green deployment strategies, ensuring zero-downtime releases across multiple environments.
- Containerized applications with Docker and deployed to Kubernetes clusters; maintained comprehensive test coverage with JUnit, Mockito, and Selenium.

Graduate Research Assistant — Computer Vision

University of Denver

September 2023 - March 2025 Remote

Supported PhD candidates on applied deep learning research projects while concurrently working full-time at Sordar Studios.

- Trained CNN-based object detection models on **datasets of 20,000+ images** across two research projects: underwater organism/debris classification and ocular disease diagnosis from eye X-ray scans (cataracts, glaucoma, and other conditions).
- Applied transfer learning and fine-tuning on pretrained architectures (**EfficientNet, ResNet, MobileNet**) with data augmentation pipelines to maximize model accuracy on limited medical and environmental datasets.
- Built end-to-end data pipelines — web scraping (Selenium, BeautifulSoup, Scrapy), cleaning and transformation (Pandas, NumPy) — to prepare large datasets for model training.
- Performed EDA, hypothesis testing, correlation analysis, and K-Means clustering using Matplotlib and Seaborn to surface insights and validate data quality before modeling.

EDUCATION

Masters of Science in Data Science
University of Denver
September 2023 – March 2025

Bachelor's in Computer Science
University of California, Santa Cruz
September 2018 – December 2020

PROJECTS

Convolutional Neural Network-Powered Food Recognition **2025**

- Data Preprocessing: Applied normalization, resizing, and format conversions to prepare fruit and vegetable images for model training and improve consistency across the dataset.
- Model Optimization: Compared custom CNN architectures and various pretrained models (e.g., EfficientNet, ResNet, MobileNet) by tuning layers, neurons, epochs, and fine-tuning strategies to identify the best setup.
- Best Model Performance: Achieved high test accuracy by selecting the best-performing model, with strong visual alignment between predicted labels and actual food images.

Spotify API: EDA - What Variables are Associated with Song Popularity **2024**

- Scraped 1,000 random song titles from random-song.com using Selenium and BeautifulSoup, then cleaned the data for consistent formatting.
- Queried Spotify's API to retrieve audio features for each song, including danceability, valence, speechiness, and other track-level attributes.
- Conducted exploratory data analysis using visualizations, correlation matrices, and KMeans clustering to identify patterns and group similar songs.